

Network Hardware & Devices

The Devices You Will Defend

Device	Layer	Function
Switch	L2 (Data Link)	Forwards frames by MAC address
Router	L3 (Network)	Routes packets by IP address between networks
Firewall	L3-L7	Filters traffic by policy
AP (Access Point)	L2-L3	Bridges wireless clients
Load balancer	L4/L7	Distributes traffic
IDS/IPS	L3-L7	Detects/prevent intrusion

MAC vs IP Addressing

- MAC address – physical, burned into NIC (48-bit hex)
- IP address – logical, assigned (32-bit for IPv4, 128-bit for IPv6)

ARP (Address Resolution Protocol) maps IP → MAC on a LAN.

ARP spoofing is the classic MITM attack: an attacker sends fake ARP replies claiming "I have the gateway's IP" so all traffic flows through them.

Network Topology and Segmentation

Modern networks use segmentation:

- DMZ (public-facing servers)

Internal LAN

Management VLAN

Guest Wi-Fi

IoT VLAN (from module 04)

Segmentation limits blast radius: a compromised web server in the DMZ cannot directly reach the finance team's VLAN without crossing a firewall.

The Spanning Tree, VLANs, and Trunks

Switches segment broadcast domains using **VLANs**. VLAN trunking (802.1Q tags) lets one physical link carry many VLANs.

Trunking attack: If a device can negotiate a trunk, it can join any VLAN – a common switch attack (DTP desync, double-tagging).

Firewalls as Network Security

Packet filter (stateless): inspects headers only

Stateful: tracks connections

NGFW: inspects at application layer, integrates IDS/IPS

WAF: protects web apps specifically

Security mindset: Network security is about *zones and policy*. An attacker's dream is a flat network where everything can reach everything. Your job is to make every zone its own fortress.

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Module complete ✓